

```
=====
STUDENT MANAGEMENT SYSTEM
=====
```

1. Add Student
2. View All Students
3. Search Student by ID
4. Search Student by Name
5. Search Students by Course
6. Update Student
7. Delete Student
8. Display Statistics
9. Display Top Performing Students
10. Sort Students
11. Course-wise Student Count
12. Save Data
13. Exit

```
=====
Enter your choice: 1
```

```
===== ADD STUDENT =====
```

Enter Student ID: STU101

Rahul Sharma

Computer Science

21

Enter Name: Enter Course: Enter Age: Enter Marks: 87.5

Student added successfully!

Grade : A

Status : PASS

Enter your choice: 2

===== ALL STUDENTS =====

Student ID : STU101
Name : Rahul Sharma
Course : Computer Science
Age : 21
Marks : 87.5
Grade : A
Status : PASS

Student ID : STU102
Name : Priya Singh
Course : Data Science
Age : 20
Marks : 94.0
Grade : A+
Status : PASS

Student ID : STU103
Name : Aman Kumar
Course : Information Technology
Age : 22
Marks : 72.0
Grade : B
Status : PASS

Student ID : STU104
Name : Ananya Verma
Course : Artificial Intelligence

```
=====
                        STUDENT MANAGEMENT SYSTEM
=====
```

1. Add Student
2. View All Students
3. Search Student by ID
4. Search Student by Name
5. Search Students by Course
6. Update Student
7. Delete Student
8. Display Statistics
9. Display Top Performing Students
10. Sort Students
11. Course-wise Student Count
12. Save Data
13. Exit

```
=====
Enter your choice:
```

```
[Auto-Save] Student data saved successfully.
```

```
3
```

```
===== SEARCH BY ID =====
```

```
Enter Student ID: STU104
```

```
-----
Student ID : STU104
```

```
Name       : Ananya Verma
```

```
Course     : Artificial Intelligence
```

```
Age        : 21
```

```
Marks      : 96.0
```

```
Grade      : A+
```

```
Status     : PASS
-----
```

```
=====
                        STUDENT MANAGEMENT SYSTEM
=====
```

1. Add Student
2. View All Students
3. Search Student by ID
4. Search Student by Name
5. Search Students by Course
6. Update Student
7. Delete Student
8. Display Statistics
9. Display Top Performing Students
10. Sort Students
11. Course-wise Student Count
12. Save Data
13. Exit

```
=====
Enter your choice: 4
```

```
===== SEARCH BY NAME =====
```

```
Enter Name: Ananya Verma
```

```
-----
Student ID : STU104
Name       : Ananya Verma
Course     : Artificial Intelligence
Age        : 21
Marks      : 96.0
Grade      : A+
Status     : PASS
-----
```

=====

1. Add Student
2. View All Students
3. Search Student by ID
4. Search Student by Name
5. Search Students by Course
6. Update Student
7. Delete Student
8. Display Statistics
9. Display Top Performing Students
10. Sort Students
11. Course-wise Student Count
12. Save Data
13. Exit

=====

Enter your choice: 5

===== SEARCH BY COURSE =====

Enter Course: Artificial Intelligence

Student ID : STU104

Name : Ananya Verma

Course : Artificial Intelligence

Age : 21

Marks : 96.0

Grade : A+

Status : PASS

```
=====
                        STUDENT MANAGEMENT SYSTEM
=====
```

1. Add Student
2. View All Students
3. Search Student by ID
4. Search Student by Name
5. Search Students by Course
6. Update Student
7. Delete Student
8. Display Statistics
9. Display Top Performing Students
10. Sort Students
11. Course-wise Student Count
12. Save Data
13. Exit

```
=====
Enter your choice:
```

```
[Auto-Save] Student data saved successfully.
```

```
8
```

```
===== STUDENT STATISTICS =====
```

```
Total Students : 5
```

```
Average Marks  : 77.50
```

```
Highest Marks   : 96.00
```

```
Lowest Marks    : 38.00
```

```
Passed Students: 4
```

```
Failed Students: 1
```

```
Unique Courses  : 4
```

```
=====
STUDENT MANAGEMENT SYSTEM
=====
```

1. Add Student
2. View All Students
3. Search Student by ID
4. Search Student by Name
5. Search Students by Course
6. Update Student
7. Delete Student
8. Display Statistics
9. Display Top Performing Students
10. Sort Students
11. Course-wise Student Count
12. Save Data
13. Exit

```
=====
Enter your choice: 9
```

```
===== TOP PERFORMERS =====
```

1. Ananya Verma | STU104 | Marks: 96.0 | Grade: A+
2. Priya Singh | STU102 | Marks: 94.0 | Grade: A+
3. Rahul Sharma | STU101 | Marks: 87.5 | Grade: A
4. Aman Kumar | STU103 | Marks: 72.0 | Grade: B
5. Rohit Gupta | STU105 | Marks: 38.0 | Grade: F

```
=====
                        STUDENT MANAGEMENT SYSTEM
=====
```

1. Add Student
2. View All Students
3. Search Student by ID
4. Search Student by Name
5. Search Students by Course
6. Update Student
7. Delete Student
8. Display Statistics
9. Display Top Performing Students
10. Sort Students
11. Course-wise Student Count
12. Save Data
13. Exit

```
=====
Enter your choice: 10
```

```
===== SORT STUDENTS =====
```

1. Marks - Highest to Lowest
2. Marks - Lowest to Highest
3. Name - A to Z
4. Student ID

Enter choice: 11

Invalid choice.

```
PS C:\miniprojectJAVA> java -cp out Main
```

```
=====
                        STUDENT MANAGEMENT SYSTEM
=====
```

1. Add Student
2. View All Students
3. Search Student by ID
4. Search Student by Name
5. Search Students by Course
6. Update Student
7. Delete Student
8. Display Statistics
9. Display Top Performing Students
10. Sort Students
11. Course-wise Student Count
12. Save Data
13. Exit

```
=====
Enter your choice: 7
```

```
===== DELETE STUDENT =====
```

```
Enter Student ID: STU105
```

```
-----
Student ID : STU105
```

```
Name      : Rohit Gupta
```

```
Course    : Computer Science
```

```
Age       : 20
```

```
Marks     : 38.0
```

```
Grade     : F
```

```
Status    : FAIL
```

```
-----
Are you sure you want to delete? (Y/N): y
```

```
Student deleted successfully!
```